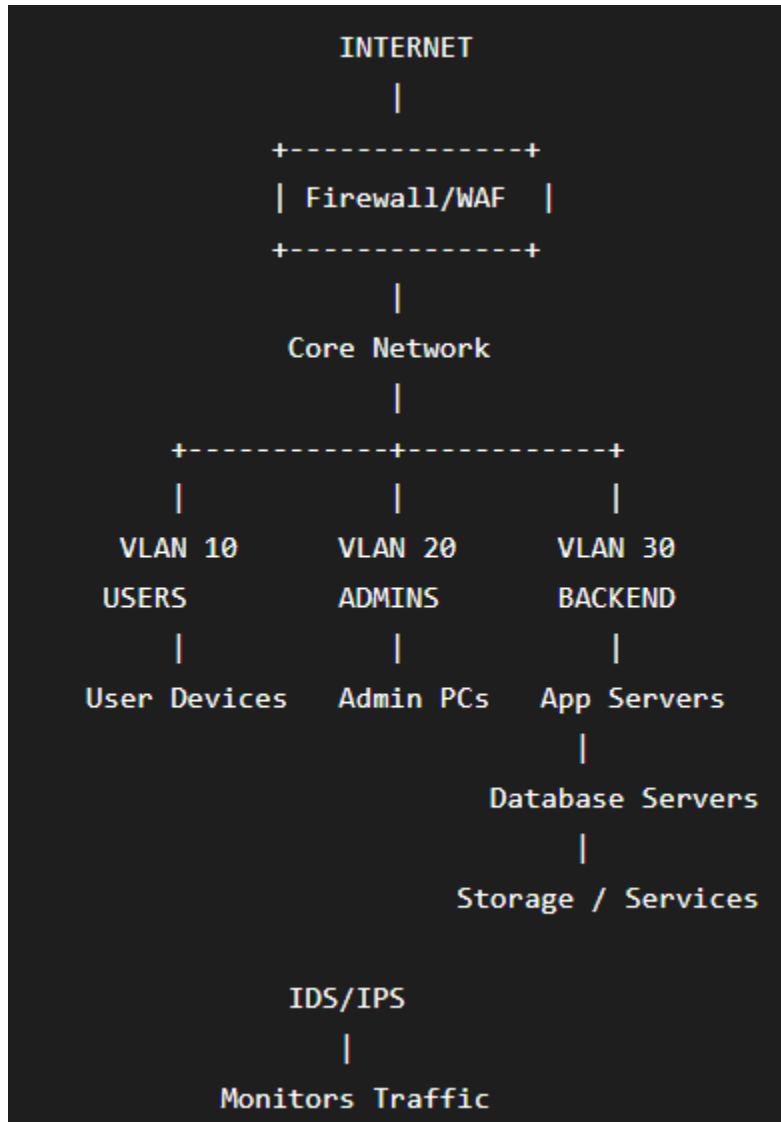


Task 1: Music Streaming Application Network Diagram

You can create this in draw.io.

Use this structure:



VLAN Design

VLAN	Purpose	Example Subnet
VLAN 10	User traffic	192.168.10.0/24
VLAN 20	Administration	192.168.20.0/24

VLAN 30	Backend services	192.168.30.0/24
---------	------------------	-----------------

Security Controls

- VLAN segmentation
- Firewall rules
- IDS/IPS monitoring
- IAM for cloud resources
- Encryption
- Restricted database access

Task 2: Playlist Read-Only IAM Policy

Example AWS IAM policy:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ReadPlaylists",
      "Effect": "Allow",
      "Action": [
        "s3:GetObject",
        "s3:ListBucket"
      ],
      "Resource": [
        "arn:aws:s3:::music-playlists",
        "arn:aws:s3:::music-playlists/*"
      ]
    },
    {
      "Sid": "DenyPlaylistModification",
      "Effect": "Deny",
      "Action": [
```

```
    "s3:PutObject",
    "s3:DeleteObject"
  ],
  "Resource": "arn:aws:s3:::music-playlists/*"
}
]
```

This allows the user to read playlist data while explicitly denying modification and deletion of playlist objects.

Task 3: IDS Detection in a Zomato-Style Platform Scenario

Suppose an attacker repeatedly sends unusual requests to a food-ordering application's login and payment APIs. The IDS detects a large number of requests from the same source IP containing suspicious patterns such as SQL injection strings or repeated failed authentication attempts.

Alert

The IDS could generate:

ALERT: Possible SQL Injection / Brute Force Activity
Source IP: 192.168.1.50
Destination: Food Ordering Web Server
Severity: HIGH

The security team can investigate the source IP, timestamps, requested URLs and payload patterns.

Task 4: Cloud Security Tool

Tool: AWS GuardDuty

For a Flipkart-style e-commerce platform, Amazon GuardDuty can be used to monitor AWS environments for suspicious activity and potential threats.

Example

If an attacker obtains compromised AWS credentials and starts making unusual API calls, GuardDuty can analyze relevant activity and generate a security finding.

Security Workflow

AWS Resources

|

v

CloudTrail / DNS / Network Activity

|

v

GuardDuty

|

v

Security Finding

|

v

Security Team Investigation

|

v

Remediation

Task 5: Secure Social Media Architecture Checklist

7+ Step Checklist

1. Segment the network using VLANs to separate users, administrators and backend services.
2. Deploy firewalls between sensitive network segments.
3. Deploy IDS/IPS to detect and potentially prevent malicious network activity.

4. Implement IAM using role-based access control and least privilege.
5. Enable MFA for administrator and privileged accounts.
6. Encrypt sensitive data both in transit and at rest.
7. Deploy cloud security monitoring such as AWS GuardDuty and CloudWatch.
8. Enable centralized logging for authentication, network and application activity.
9. Regularly patch systems and scan for vulnerabilities.
10. Create backups and incident-response procedures to recover from attacks.